

NVG EM-Response Derivation: From NJL Chiral Condensate to Dense-Matter Gauge-Kinetic Renormalisation

Companion note to NVG Magnetar Mechanism V2

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Repository: <https://github.com/infosave2007/vmf>

Purpose

The V2 magnetar preprint of the Nonlinear Vacuum Gravity (NVG) programme introduces a structural amplification factor $\Gamma_{\text{struct}} \sim 2.2\text{--}3.3$ that sits on top of the standard flux-freezing compression and is held responsible for converting an Of?p/Bp-class progenitor field $B_{\star,\text{surf}} \sim 10^3\text{--}10^5$ G into a magnetar dipole field $B \sim 10^{14}\text{--}10^{15}$ G. In V2 this factor was *motivated* by a vacuum-/order-parameter-dependent gauge kinetic term but not derived from standard microphysics. The present note closes that gap by performing the controlled microphysical derivation explicitly, with two parallel calculations implemented in `verification/nvg_em_response_derivation.py`.

The headline outcome is intentionally honest:

1. Standard one-loop QED-in-medium vacuum polarisation, evaluated with the NJL constituent quark mass solved from the textbook gap equation, gives $\Gamma_{\text{struct}}^{(\text{QED})} = 1.000\text{--}1.004$. The perturbative channel falls short of the magnetar benchmark by **two orders of magnitude**. There is no QED-only way to obtain a factor of two from a one-loop log.
2. A non-perturbative coupling between the chiral condensate $\sigma(\rho)$ and the gauge kinetic term, of the natural-dimensional-analysis form

$$Z_\chi(\sigma) = 1 - \lambda (1 - \sigma^2(\rho)/\sigma^2(0)),$$

reproduces the V2 benchmark band with a single dimensionless coupling $\lambda \simeq 0.88\text{--}1.01$ at twice nuclear saturation density.

The previously “motivated” V2 step is therefore replaced by an explicit benchmark parametrisation: if the magnetar amplification is encoded in a condensate-driven gauge-kinetic response of the form above, the required coupling is a single number λ of order unity. This number is, in principle, testable by lattice QCD and by chiral-effective-field-theory calculations of the photon two-point function as a function of $\langle \bar{q}q \rangle$. If those calculations give $\lambda \ll 1$, the minimal-QCD version of the NVG magnetar channel is falsified.

1. NJL Background

We use the standard two-flavour Nambu–Jona-Lasinio model with a three-momentum cutoff, in the conventions of Klevansky (1992) and Buballa (2005):

Parameter	Value
Cutoff Λ	631.3 MeV
Coupling $G \Lambda^2$	2.14
Current mass m_q	5.5 MeV

The gap equation for the constituent mass at chemical potential μ_q and temperature T is

$$M_q = m_q + 4G N_c N_f M_q \int_0^\Lambda \frac{k^2 dk}{2\pi^2 E_k} [1 - n_F(E_k - \mu_q) - n_F(E_k + \mu_q)], \quad E_k = \sqrt{k^2 + M_q^2}.$$

The script solves this by bisection of the residual on $M_q \in (m_q, \Lambda]$, taking the largest (broken-symmetry) root when one exists.

Numerical vacuum solution:

$$M_q(0, 0) = 317.7 \text{ MeV}, \quad |\langle \bar{q}q \rangle|^{1/3} = 244.0 \text{ MeV},$$

in agreement with textbook values (~ 313 MeV and ~ 250 MeV respectively).

2. Channel (A): Perturbative QED-in-Medium

The standard one-loop contribution of a charged Dirac fermion of mass M to the gauge kinetic term, in the on-shell scheme with the EFT cutoff Λ playing the role of the UV scale, is

$$Z_{\text{QED}}(M) = 1 - \frac{\alpha N_c}{3\pi} \left(\sum_q Q_q^2 \right) \ln \frac{\Lambda^2}{M^2}, \quad \sum_q Q_q^2 = \frac{4}{9} + \frac{1}{9} = \frac{5}{9} \text{ (for } u, d\text{)}.$$

With the standard flux-freezing argument applied to the *physical* magnetic field, the structural amplification on top of geometric compression is

$$\Gamma_{\text{struct}}^{(\text{QED})}(\rho, T) = \sqrt{\frac{Z_{\text{QED}}(M_q(0, 0))}{Z_{\text{QED}}(M_q(\rho, T))}}.$$

Evaluated over the proto-neutron-star corridor $\rho \in [1, 5] n_0$, $T \in [0, 50]$ MeV, the script returns

$$\Gamma_{\text{struct}}^{(\text{QED})} \in [1.0004, 1.0037].$$

This is the honest reading of one-loop QED. The prefactor $(\alpha N_c \sum Q^2)/(3\pi) \approx 1.3 \times 10^{-3}$ multiplied by the modest $\log \ln(\Lambda^2/M_q^2) \sim 6$ at $M_q \sim 20$ MeV gives a 1% effect at most. **One-loop QED in medium cannot produce a factor-of-two amplification.**

The implication for the magnetar paper is structural, not numerical: the amplification claimed in V2 was always implicitly a *non*-perturbative QCD-vacuum effect, and the present derivation makes that explicit.

3. Channel (B): Non-Perturbative Condensate-Photon Coupling

The natural QCD-scale effective Lagrangian that captures a chiral-condensate dependent renormalisation of the photon kinetic term is

$$\mathcal{L}_{\text{eff}} = -\frac{Z_\chi(\sigma)}{4} F_{\mu\nu} F^{\mu\nu}, \quad Z_\chi(\sigma) = 1 - \lambda \left(1 - \frac{\sigma^2(\rho)}{\sigma^2(0)}\right),$$

where $\sigma(\rho) \equiv \langle \bar{q}q \rangle(\rho)$ is the chiral order parameter and λ is a dimensionless coupling whose scale is set by QCD ($G_{\text{NJL}} \sim \text{GeV}^{-2}$, *not* by α). Naive dimensional analysis predicts $\lambda \sim \mathcal{O}(0.5 - 1)$.

The amplification factor is

$$\Gamma_{\text{struct}}^{(\chi)}(\rho) = \frac{1}{\sqrt{Z_\chi(\rho)}} = \frac{1}{\sqrt{1 - \lambda(1 - r^2(\rho))}} \quad \text{with} \quad r(\rho) = \sigma(\rho)/\sigma(0).$$

Inverting for λ at a given target Γ_{struct} :

$$\lambda(\rho; \Gamma_{\text{struct}}) = \frac{1 - \Gamma_{\text{struct}}^{-2}}{1 - r^2(\rho)}.$$

Tabulated values (from the script, $T = 10$ MeV throughout):

ρ/n_0	$r(\rho) = \sigma/\sigma_0$	$\lambda(\Gamma = 2.2)$	$\lambda(\Gamma = 3.3)$
1.0	0.801	2.21	2.53
1.5	0.491	1.04	1.20
2.0	0.312	0.88	1.01
2.5	0.195	0.82	0.94
3.0	0.145	0.81	0.93
5.0	0.081	0.80	0.91

The behaviour is monotonic and physically transparent. At low density $\rho \lesssim n_0$ partial chiral restoration is still mild, so the linear ansatz needs $\lambda > 1$ to deliver the required amplification — i.e. the EFT is being pushed to the edge of its validity. From $\rho \gtrsim 1.5 n_0$ the condensate has dropped enough that the required λ is well below unity, and converges to the asymptotic value $\lambda_\infty(\Gamma = 2.2) \simeq 0.79$, $\lambda_\infty(\Gamma = 3.3) \simeq 0.91$ once $r \rightarrow 0$ (full chiral restoration).

4. Headline at $\rho = 2 n_0$

The benchmark proto-magnetar interior density is $\rho \simeq 2 n_0$, where the chiral condensate has fallen to $r = 0.31$ of its vacuum value. There the condensate coupling required to reproduce the V2 benchmark band $\Gamma_{\text{struct}} \in [2.2, 3.3]$ is

$$\lambda \in [0.88, 1.01].$$

This is order unity, sits inside the naive-dimensional-analysis window, and defines a single benchmark number that lattice QCD could in principle target through the photon two-point function as a function of $\langle \bar{q}q \rangle$ at finite μ_q . The companion derivation therefore converts the V2 amplification claim from an unlabelled postulate into an explicit EFT target, but not yet into a first-principles QCD prediction.

5. What This Closes and What It Does Not

What this note **does** close:

- The structural objection that the EM response in V2 was “motivated, not derived” from standard microphysics at the level of an explicit EFT target. The derivation chain is now explicit: NJL gap equation $\rightarrow$ chiral order parameter $\sigma(\rho) \rightarrow$ effective gauge kinetic term $Z_\chi(\sigma) \rightarrow$ amplification factor $\Gamma_{\text{struct}}(\rho; \lambda)$. Standard QED accounts for $\sim 1\%$; the rest is carried by λ , which is a single named non-perturbative parameter.
- The implicit assumption in V2 is now explicit: the magnetar mechanism requires $\lambda \sim 1$ at $\rho \sim 2n_0$. This is an order-of-magnitude prediction with a clean falsifier (lattice).

What this note does **not** close:

- The first-principles lattice calculation of λ itself. We give the required value and the formula; we do not derive λ from QCD. That remains future work for the lattice community.
- The detailed form of $Z_\chi(\sigma)$ beyond the leading $(1 - r^2)$ term. Higher-order terms in the EFT may shift λ by $\mathcal{O}(20\%)$ near $r \sim 1$; they are negligible at $r \lesssim 0.3$ where the headline value lives.

5a. Standard-QCD Upper Bound on λ (Addendum)

Section 4 derived the *required* coupling $\lambda \in [0.88, 1.01]$ at $\rho = 2n_0$. A natural follow-up question is: what does standard QCD itself predict for λ , by aggregating every known operator that couples $\langle \bar{q}q \rangle$ (or its functional surrogates) to the photon kinetic term? The companion script `verification/nvg_em_response_higher_order.py` performs exactly this aggregation, channel by channel, from textbook constants alone.

The channels included are:

1. **(A) One-loop quark Z_QED** with the in-medium NJL constituent mass. This is the perturbative contribution of Section 2, reproduced here.
2. **(B) Charged-pion one-loop contribution** with in-medium $m_\pi(\rho)$ from the Gell-Mann–Oakes–Renner relation and Brown–Rho scaling of $f_\pi(\rho)$. Effect at any density: $|\delta Z_\pi| \lesssim 10^{-4}$.
3. **(C) Vector-meson-dominance γ – ρ^0 mixing**, evaluated at the neutron-star coherence scale $Q \sim 1/R_{\text{NS}}$. For a quasi-static magnetar field this is essentially zero; we include it to set the upper bound rigorously.
4. **(D) Magnetic susceptibility of the QCD vacuum** $\chi_{\text{QCD}} \langle \bar{q}q \rangle$ from the operator $\langle \bar{q} \sigma_{\mu\nu} q \rangle_F = \chi \langle \bar{q}q \rangle e_q F_{\mu\nu}$. We use the lattice value $\chi_{\text{QCD}} = -2.08 \text{ GeV}^{-2}$ (Bali et al, JHEP 2012) as the best estimate and the SVZ value -3.15 GeV^{-2} (Belyaev–Kogan 1984) as the most generous upper bound. The dimensional combination that enters δZ is $|\chi| m_q (\sum_q Q_q^2) |\langle \bar{q}q \rangle(0)| (1 - r)/\Lambda_\chi^2$ with $\Lambda_\chi = 4\pi f_\pi$.

Aggregated standard-QCD prediction at $\rho = 2n_0$ (under the same ansatz $Z_\chi = 1 - \lambda(1 - r^2)$):

$$\lambda_{\text{std}}(\text{best}) \simeq 3.9 \times 10^{-3}, \quad \lambda_{\text{std}}(\text{most generous}) \simeq 4.0 \times 10^{-3}.$$

The required value for the V2 magnetar amplification is $\lambda \sim 0.9$. The shortfall is therefore

$\lambda_{\text{req}}/\lambda_{\text{std}} \sim 220.$

Including next-to-leading-order EFT corrections $Z = 1 - \lambda(1 - r^2) - \mu(1 - r^2)^2$ with the conservative natural-size estimate $|\mu| = 0.5 |\lambda|$ reduces the required λ by $\sim 30\%$ (to ~ 0.6), which does not change the conclusion: standard QCD operators fall short by **roughly two orders of magnitude**.

5b. Honest Interpretation

This finding is a real physical result that must be reported, not buried. Two scientifically honest options follow.

(a) Lower the V2 claim. Accept $\lambda \sim 4 \times 10^{-3}$ as the standard-QCD answer. Then the NVG structural amplification factor is

$$\Gamma_{\text{struct}}^{(\text{std QCD})} \simeq \sqrt{1/(1 - 4 \times 10^{-3})} \approx 1.002,$$

i.e. about 0.2%. In this reading, the NVG channel provides a small correction on top of the conventional Thompson–Duncan dynamo or Ferrario–Wickramasinghe fossil-field mechanisms, not the dominant amplification. The magnetar-paradox resolution then *cannot* be carried by the NVG sector alone, and the V2 corridor coverage statistics (32/32 sources under the magnetic-progenitor prior) is recovered only because the prior itself already does most of the work.

(b) Keep the V2 claim, but identify the missing operator. The factor of ~ 200 shortfall cannot come from any tweak of Standard Model parameters. It requires a new operator coupling the photon kinetic term to a dense-matter order parameter, with strength set by the NVG sector rather than by χ_{QCD} . The cleanest concrete realisation is a hidden $U(1)$ field A'_μ that mixes with the visible photon via kinetic mixing $-\frac{1}{2}\varepsilon F_{\mu\nu}F'^{\mu\nu}$, with the mixing parameter ε controlled by the NVG order parameter $\mathcal{W}$ (or $M_\Omega(n_B)$). This is *Standard Model + one new vector + one mixing operator*, fully consistent with the gauge-symmetry structure of the NVG framework, but it must be acknowledged as **new physics in the gauge sector**, not a derivation from standard QCD vacuum response.

Either choice is publishable. Conflating them is not. The present companion note treats the V2 magnetar amplification as option (b) by construction – the NVG sector is, by definition, beyond minimal QCD – and the value $\lambda \sim 0.9$ is a constraint on the NVG-induced kinetic-mixing parameter, not a prediction of QCD. One illustrative completion of this type, based on density-activated dark-photon kinetic mixing, is analysed separately in `article/NVG_DARK_PHOTON_PREPRINT.md`; that analysis should be read as an exploratory benchmark, not as a full exclusion-level phenomenology study. The numerical script here only makes the QCD shortfall visible and quantitative.

6. Reproducing the Numbers

`python3.12 verification/nvg_em_response_derivation.py`

The script prints the full scan and the headline. The numerical output is also frozen in `verification/nvg_em_response_derivation_output.txt`.

References

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